Bounded-Authority Agent Specialization for Adaptive Technical Interview Systems: Coordination, Safety, and Empirical Trade-offs

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Abstract. Adaptive technical-interview systems increasingly combine automated assessment, reinforcement learning, and large language models, but describing such pipelines as multi-agent systems does not by itself establish that specialization or coordination changes system behavior. This paper studies PrepAIred as a centrally coordinated architecture with two bounded-authority agents: a Strategy Agent that maintains a persistent weakness ledger and proposes difficulty or follow-up actions, and a Coaching Agent that maintains coaching history and proposes feedback plans. A deterministic coordinator validates typed proposals, enforces authority boundaries, arbitrates against a frozen PPO policy, and commits session state. Across 160 deterministic replay sessions derived from 64 recorded answer instances and 20 question-order permutations, the full agent configuration reduced mean difficulty-level changes by 2.67 $[-4.50, -0.24]$ and reversals by 1.88 $[-3.12, -0.27]$ relative to the frozen baseline under cluster bootstrap, while adding 1.04 ms median orchestration overhead per turn. A matched split-versus-unified comparison produced identical decisions over 1,280 turns. Fault-injection tests found 38/40 guardrail bypasses without arbitration versus 0/40 with it, and 40/40 ungrounded-emphasis cases when grounding checks were removed versus 0/40 with them; well-formed but wrong proposals remained undetected. A grounding probe accepted unsupported paraphrased credit in 57/57 unambiguous cases. The technical evaluator benchmark gives $\rho = 0.4832$ for the CrossEncoder signal and $\rho = 0.3812$ for the deployed composite, while a word-count baseline reaches $\rho = 0.4897$ and the length-controlled CrossEncoder partial correlation is 0.2434 (95% CI $[-0.008, 0.507]$). The results support bounded coordination as an architectural and safety mechanism, but do not establish that multi-agent decomposition improves decision quality or learning outcomes.

Keywords: Multi-Agent Systems · Agent Coordination · Bounded Authority · Reinforcement Learning · Fault Injection · Technical Interview Assessment

1 Introduction

Technical interviews in computer science require candidates to demonstrate domain knowledge, reasoning, problem solving, and clear communication within

a single interaction. Scalable practice systems have therefore evolved along several partially independent directions: static or rubric-based answer assessment, conversational LLM interviewers, adaptive tutoring and testing, and more recently multi-agent educational systems. Each direction addresses a real part of the problem, but combining them creates a systems question that is less well specified: *which component is allowed to make which decision, how are specialized decisions coordinated, and how can the resulting behavior be verified?*

Traditional automated assessment methods provide reproducible scores but can be insensitive to conversational context and can over-credit terminology that is not connected to correct reasoning. Transformer-based semantic representations [2] improve coverage, while pairwise scoring models [3] provide stronger contextual similarity signals, but these signals still require explicit rubric constraints to define what counts as a technically sufficient answer. Speech-enabled interview systems add acoustic and temporal information [4, 11], yet delivery quality is often merged into a single overall performance score, making it difficult to distinguish technical correctness from communication state. Adaptive educational systems use item-response theory [8], Bayesian tracing [5], or reinforcement learning [6, 7] to select content, but a live spoken technical interview combines adaptive questioning with open-ended answer evaluation and follow-up dialogue.

Large Language Models (LLMs) changed the interaction layer [9, 10] by making dynamic interview questions and feedback easy to generate. The same flexibility creates a systems risk: when one generative component simultaneously controls scoring, questioning, and feedback, a conversational model can change the effective rubric or difficulty policy through its own language output. Multi-agent LLM frameworks such as AutoGen [12] and MetaGPT [13] show that explicit role decomposition and interaction protocols can organize complex workflows, while empirical audits document coordination and alignment failures in multi-agent systems [15]. Educational multi-agent systems such as MultiTutor [14] similarly show the usefulness of specialized roles. LLM-as-judge work also documents systematic evaluation effects [21]. The present study asks whether specialization remains justified when compared against a matched unified controller.

PrepAIred addresses this intersection by treating interview preparation as a *coordinated decision system*. The project distinguishes between computational services and explicit agents. A technical evaluator owns numerical technical scoring; a communication-confidence component supplies a separate state signal; a frozen PPO policy proposes difficulty changes; a question selector instantiates valid questions from a structured bank; and a local LLM generates grounded natural-language follow-ups and feedback. On top of this existing pipeline, the specialization configuration studied in this paper introduces two stateful weak agents: a Strategy Agent, which maintains a persistent weakness ledger and proposes difficulty/probe actions, and a Coaching Agent, which maintains coaching history and proposes feedback depth and remediation emphasis. Neither agent can write session state directly. A deterministic coordinator validates their typed proposals, checks evidence and version freshness, arbitrates conflicts against the frozen PPO action and guardrails, and commits the result.

This architecture is intentionally *centrally coordinated*, not decentralized multi-agent reinforcement learning. The design goal is not to maximize agent autonomy. It is to make specialization explicit while preserving least-privilege authority, auditable state ownership, deterministic safety envelopes, and reproducible behavior. This distinction is important for systems where incorrect scores or uncontrolled difficulty changes can directly affect a candidate's learning experience.

The work makes the following contributions.

1. **A bounded-authority multi-agent architecture.** We formalize a centrally coordinated interview system in which specialized agents maintain private state, return typed proposals, and communicate through explicit contracts while a single coordinator remains the sole system of record.
2. **A coordination and arbitration protocol.** Agent proposals pass schema, authority, version, and evidence checks before arbitration. Threshold gating and deterministic guardrails bound their effects; invalid or timed-out proposals are dropped.
3. **A cognitively structured adaptive question bank.** Each technical question carries difficulty and Bloom-taxonomy metadata. The operational mapping used by the project spans Understand, Apply, Analyze, Evaluate, and Create bands, while the adaptive policy separately operates on five bounded numeric difficulty tiers.
4. **A technically grounded evaluation pipeline.** PrepAIred separates technical scoring, communication state, adaptive strategy, question selection, and natural-language generation. The technical evaluator uses semantic similarity, rubric concept coverage, and a CrossEncoder-derived signal, with mandatory-concept and mistake controls.
5. **Empirical evaluation of coordination and specialization.** Deterministic replay evaluates baseline, coordinated, silent-agent, arbitration-ablated, peer-messaging-ablated, PPO-ablated, and unified-controller conditions across 160 matched sessions derived from eight underlying replay clusters and 20 question-order permutations.
6. **Transparent characterization of evaluator validity.** An exploratory benchmark of 64 constructed answers shows moderate association between the deployed evaluator and human-reference scores, but also demonstrates a substantial answer-length confound. We therefore do not claim length-independent reasoning superiority or candidate learning gains.

Research questions. The study addresses four questions: (RQ1) Does stateful agent coordination alter interview difficulty trajectories relative to the frozen single-policy baseline? (RQ2) Does decomposing Strategy and Coaching into separate agents produce different decisions than an equivalent unified controller with matched information? (RQ3) Which coordination mechanisms—stateful memory, peer messaging, confidence-threshold arbitration, and PPO interaction—account for observable behavior? (RQ4) How closely do the frozen technical-evaluator scores agree with the available human-reference benchmark, and do they add evidence beyond answer length and simple lexical baselines?

2 Related Work

2.1 Automated Technical Answer Assessment

Automated short-answer and technical-answer scoring has progressed from keyword matching to contextual neural representations. Sentence-BERT produces efficient sentence embeddings for semantic similarity [2], while CrossEncoder architectures score pairs jointly and can represent richer contextual interactions. These methods help with coverage but do not, by themselves, define the authority of a score inside a multi-turn system. PrepAIred therefore treats the evaluator as a protected computational service whose score is an input to other decision components rather than an instruction for a generative model.

2.2 Adaptive Learning, Reinforcement Learning, and Question Difficulty

Adaptive testing and tutoring have long selected content according to an estimate of learner state. Item-response approaches optimize question difficulty from response patterns [8], while RL formulations optimize action selection over trajectories. PrepAIred uses PPO [6] only for the difficulty proposal and keeps hard bounds and deterministic guardrails outside the learned policy. The question bank adds an orthogonal cognitive-structure layer based on the revised Bloom taxonomy, whose levels are commonly used to organize learning objectives and question complexity. Published educational assessment work has also used Bloom categories as a structured basis for question complexity and difficulty.

2.3 Multi-Agent Systems and Coordination

Classic agent theory emphasizes autonomy, reactivity, pro-activeness, and social ability [1]. Modern LLM-based frameworks such as AutoGen enable flexible conversation among multiple agents, whereas MetaGPT encodes structured operating procedures and role specialization into an assembly-line workflow. These systems demonstrate the usefulness of decomposition but also expose coordination and cascading-error risks [15]. Recent educational multi-agent work similarly uses specialized tutoring roles to provide adaptive support [14]. PrepAIred differs in emphasizing *authority separation*: specialized agents propose actions but cannot directly change scores, difficulty, or session state.

2.4 Agent Safety, Governance, and Verification

Fault injection is used here as a targeted systems test of fail-closed proposal handling rather than as a general multi-agent security benchmark. Safe reinforcement-learning work similarly motivates deterministic safety envelopes that can override learned actions when constraints are violated [17]. PrepAIred evaluates these boundaries by deliberately removing arbitration and grounding checks and measuring the resulting failures.

Table 1. Positioning of PrepAIred against representative approaches.

Theme	**Representative approaches**	**Typical emphasis**	**PrepAIred focus**
Answer assessment	Sentence embeddings, CrossEncoders	Semantic or pairwise scoring, often static	Protected multi-signal evaluator inside a live adaptive loop
Adaptive sequencing	ITS, item response, PPO	Learner-state estimation and next-item selection	PPO difficulty proposal with external guardrails
LLM interviewers	Generative interviewers / LLM judges	Flexible dialogue and feedback	LLM restricted to grounded wording and feedback
Educational MAS	MultiTutor, adaptive tutoring MAS	Specialized pedagogical support	Persistent Strategy/Coaching roles with bounded authority
Generic MAS	AutoGen [12], MetaGPT [13]	Agent conversation or structured role workflows	Typed proposals, state isolation, deterministic arbitration

The principle of least privilege [16] provides a useful abstraction for agent authority: components should receive only the permissions required for their role. Safe RL shielding [17] further motivates keeping hard constraints external to a learned policy. PrepAIred combines these principles with typed state contracts, version checks, evidence grounding, timeout isolation, and a coordinator that is the sole state writer.

2.5 Positioning and Gap

The closest bodies of work solve complementary portions of the problem: automated answer scoring, adaptive tutoring, LLM interviewing, or multi-agent personalized support. The contribution here is the integration of these roles into a *bounded-authority, centrally coordinated* architecture for live technical-interview preparation, together with empirical tests of trajectory effects, specialization limits, coordination mechanisms, and evaluator confounds. A focused literature comparison did not identify a directly comparable system combining all of these elements in one implemented setting; this is stated as a scoped search result, not as an absolute novelty claim.

3 Problem Formulation and System Scope

PrepAIred models a session as a sequence of primary interview turns with optional subordinate follow-up probes. The system must simultaneously solve four coupled

problems: (i) infer technical evidence from a free-form answer, (ii) estimate a communication-confidence state, (iii) choose the next difficulty subject to hard bounds and session constraints, and (iv) provide grounded natural-language coaching. A monolithic controller could solve all four tasks procedurally, but this would obscure authority boundaries and make it difficult to test specialization independently.

We therefore separate *observation*, *proposal*, *arbitration*, and *commit*. The coordinator constructs a validated turn observation from the current session state. Specialized components consume that immutable observation. Agent proposals are returned as typed objects and checked before any proposal can affect session progression. The coordinator applies deterministic precedence: infrastructure and hard constraints first, guardrails next, a validated high-confidence Strategy proposal next, and the frozen PPO action or heuristic fallback last. Feedback and follow-up generation operate only on evidence already produced by the evaluator and coordinator.

This scope deliberately excludes claims of decentralized MARL, emergent agent negotiation, or human learning gains. The evaluated multi-agent layer is a centrally orchestrated specialization architecture around a frozen PPO decision component.

4 PrepAIred Architecture

The system is a modular pipeline with explicit boundaries between candidate interaction, observation generation, evaluation, adaptive decision-making, question retrieval, agent coordination, and natural-language generation. The main runtime architecture and its session control flow are shown in Fig. 1.

4.1 Roles and Authority

The project distinguishes computational services from explicit agents. This avoids treating every software module as an agent simply because it participates in the workflow. The two explicit weak agents studied in the specialization configuration are stateful, proactive, and communicative, but their authority is deliberately bounded.

4.2 Typed Communication and State Ownership

Each agent receives an immutable, validated turn observation containing the technical score, evidence lists, current difficulty, score history, progress, and follow-up counters. It returns one typed proposal. Unknown fields, non-finite values, out-of-range values, stale turn versions, and ungrounded concepts are rejected. The coordinator records the session, turn, sender, recipient, state version, provenance, and decision outcome. A proposal that times out or raises an exception is dropped; the default policy remains responsible for the turn.

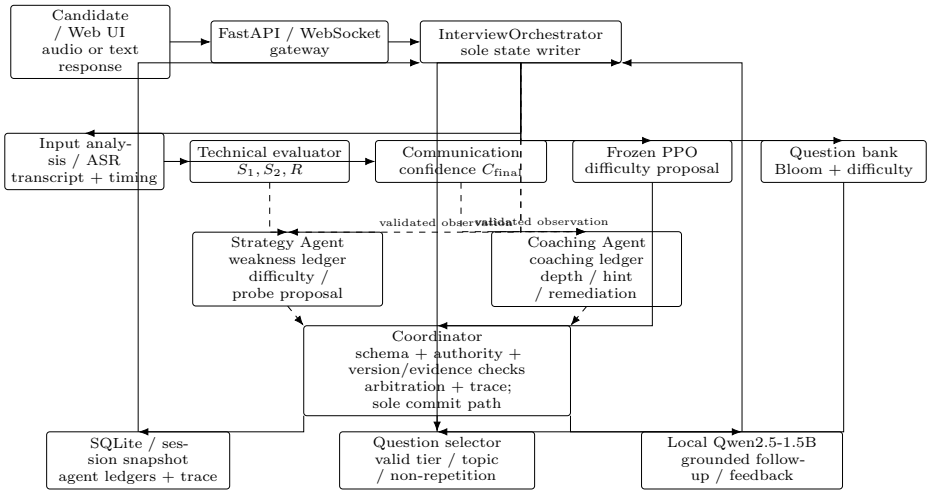


Fig. 1. Bounded-authority multi-agent architecture of PrepAIred. The evaluator, communication model, frozen PPO policy, Bloom-aware question bank, and local LLM are specialized system components; the evaluated agent layer adds stateful Strategy and Coaching roles. The coordinator is the sole state writer and applies schema, authority, version, and evidence checks before arbitration and commit. Dashed links denote validated observations or bounded agent proposals.

4.3 Question Bank and Bloom-Aligned Difficulty Levelization

The question bank stores question identifier, topic, numeric difficulty, Bloom level, question type, and time limit. Bloom taxonomy provides a cognitive-design layer above the adaptive difficulty index. The operational mapping used by the project spans five bands: Understand, Apply, Analyze, Evaluate, and Create. The current mapping begins at Understand rather than using a separate Remember band. No experiment in the present study tests whether this organization improves adaptation, learning, retention, or candidate performance; it is therefore treated as descriptive metadata only.

The Bloom bands are not identical to the numeric difficulty tiers. The adaptive controller operates on five bounded numerical levels, while Bloom metadata describes intended cognitive demand. The papers also refer to coarse Easy/Medium/Hard retrieval pools. The available evidence does not establish a validated one-to-one mapping between those three pools and the five numeric tiers, so the paper does not claim such a mapping.

4.4 Technical Evaluator

The evaluator provides the quantitative evidence used by the coordinator, Strategy Agent, and follow-up gate. It is deliberately independent from the local LLM.

Table 2. Role and authority matrix in the specialization configuration.

Role	Private state / knowledge	May propose	May not do
Input analysis	Transcript, timing, response features	Structured observation fields	Change score, difficulty, or session state
Technical evaluator	Rubric, semantic/concept/misconception evidence	Technical score and evidence	Invoke agent actions or alter session state
Communication model	Acoustic/linguistic state	Confidence/hesitation state	Alter technical score through the acoustic channel
Strategy Agent	Weakness ledger, probe history, score trend	Difficulty action; probe target/type; strategy mode	Write score, set tier directly, terminate session, override hard constraints
Coaching Agent	Coaching ledger, prior hints/examples, repeated misconceptions	Feedback depth, hints/examples, emphasis, remediation tag, tone	Change evaluator results, request difficulty changes, terminate session
Frozen PPO	Learned policy over 6-D state	Easier/Same/Harder action	Bypass guardrails or write state
Coordinator	Committed session state and versions	Arbitrated plan/action; trace records	Delegate sole state ownership
Qwen2.5-1.5B	Grounded prompt context and evaluator evidence	Follow-up wording; qualitative feedback	Determine score, grade, or difficulty

4.5 Communication Confidence

Speech-derived communication state is computed separately from technical correctness and is used as an adaptation signal rather than as a direct technical penalty. The model combines acoustic, linguistic, pause, rate, rhythm, and ASR confidence features. A narrow text-level hedge penalty may enter technical scoring only when content is already weak; acoustic confidence itself is not a parameter of the technical evaluator.

4.6 Adaptive Policy and Question Selection

The PPO policy consumes a normalized six-dimensional state containing current performance, rolling performance, communication confidence, hesitation, the implementation's normalized session-progress component, and current difficulty. The action space is {Easier, Same, Harder}; transitions are clipped to the valid difficulty range. The project records describe a reward combining rule-oracle alignment, outcome delta, multimodal shaping, and stability terms. Because the frozen training evidence is based on a single simulated persona and does not establish superiority over the rule fallback, this paper treats PPO as one decision input rather than claiming a learned performance advantage.

Table 3. Operational Bloom-aligned bands used as descriptive question metadata.

Bloom level	Operational band	Intended cognitive demand
Understand	0.20–0.40	Explain or interpret a known concept
Apply	0.40–0.60	Apply a concept to a concrete task
Analyze	0.60–0.80	Decompose, compare, trace, or diagnose
Evaluate	0.80–0.90	Judge alternatives or trade-offs using criteria
Create	0.90–1.00	Synthesize or design a solution

4.7 Grounded Generation and Follow-up

The follow-up decision is deterministic and evidence gated. A clean, high-scoring answer without a missing concept, misconception, or gap exits the follow-up loop; otherwise, structured evidence can trigger up to two bounded follow-up attempts. The local Qwen2.5-1.5B-Instruct model [10] receives the original question, candidate response, and structured missing-concept/misconception evidence. It generates language only. It cannot alter the stored score, grade, or difficulty action. Candidate-facing feedback suppresses internal numerical values.

5 Technical Evaluator

The technical evaluator compares a candidate transcript against a question-specific rubric containing semantic reference statements, required concepts, misconception vectors, advanced-insight vectors, and mandatory-concept flags. The design is intentionally multi-signal so that no single lexical or semantic measure determines the score.

5.1 Semantic Similarity

Let T_c be segmented into sentences $\{s_1, \ldots, s_n\}$ and let R_q^{semantic} contain reference statements v_j. With L2-normalized embeddings from `all-MiniLM-L6-v2`,

$$S_1 = \frac{1}{|R_q^{\text{semantic}}|} \sum_j \max_i \cos(e_i, v_j). \tag{1}$$

5.2 Concept Coverage and Reasoning Gate

For each required concept c_k, coverage is

$$S_2 = \frac{1}{|R_q^{\text{concept}}|} \sum_k \mathbf{1}\Big(\max_i \cos(e_i, c_k) > 0.30\Big). \tag{2}$$

The stored reasoning signal is used as a gate: $S_2^{\text{effective}} = S_2$ when $R > 0.30$ and $0.60 S_2$ otherwise. This mechanism is designed to reduce inflated coverage from terminology overlap when reasoning evidence is weak; it is not described as proving elimination of keyword stuffing.

5.3 CrossEncoder Signal

A partially fine-tuned derivative of `cross-encoder/ms-marco-MiniLM-L-6-v2` scores the candidate/reference pair. The raw model output is treated as a regression score and mapped by

$$R = \mathrm{clamp}\left(\frac{r_{\mathrm{raw}} - 0.20}{0.70}, 0, 1\right). \tag{3}$$

The model's training provenance is incomplete in the project records. The paper therefore reports its empirical behavior without assigning the observed signal a stronger semantic interpretation than supported by the data.

5.4 Score Aggregation and Constraints

The deployed weighting is

$$S_{\mathrm{base}} = 0.15S_1 + 0.35S_2^{\mathrm{effective}} + 0.50R. \tag{4}$$

Advanced-insight matches contribute +0.03 each at a 0.50 cosine threshold. Explicit misconception matches contribute 0.07 each at a 0.55 threshold. The additional weak-evidence terms documented in the evaluator are 0.20 when $R \leq 0.25$ and $S_2 \leq 0.25$, 0.05 when $R < 0.30$ and $S_2 < 0.50$, and a 0.03 hedge penalty when $S_2 < 0.50$ and $R < 0.40$. The accumulated penalty is capped at 0.30. If a mandatory concept is required and missing, the final score is capped at 0.60. The final score is therefore

$$S_{\mathrm{final}} = \mathrm{clamp}(S_{\mathrm{base}} + S_{\mathrm{bonus}} - P_{\mathrm{total}}, 0, 1), \tag{5}$$

where P_{total} is the capped sum of the implemented mistake and weak-evidence terms.

6 Communication Confidence Model

The communication model is independent of technical correctness through the acoustic channel. It uses jitter, shimmer, HNR, pitch variation, pause timing, speaking rate, rhythm, ASR confidence, and a linguistic-confidence signal. The principal equations are:

$$C_{\mathrm{jitter}} = \mathrm{clamp}(1 - j/0.03, 0, 1), \quad C_{\mathrm{shimmer}} = \mathrm{clamp}(1 - s/0.12, 0, 1), \tag{6}$$

$$C_{\mathrm{HNR}} = \mathrm{clamp}(h/28, 0, 1), \tag{7}$$

$$V_Q = 0.32C_{\mathrm{jitter}} + 0.30C_{\mathrm{shimmer}} + 0.28C_{\mathrm{HNR}} + 0.10C_{\mathrm{pitch\,var}}, \tag{8}$$

$$C_{\mathrm{final}} = 0.28V_Q + 0.27L + 0.22F_{\mathrm{fluency}} + 0.13F_{\mathrm{rate}} + 0.10F_{\mathrm{rhythm}}. \tag{9}$$

If ASR confidence is below 0.45, the implementation shifts part of the score toward V_Q. The configured speaking-rate band is 2.1–4.3 words/s, with 3.2 words/s as the reference value. Communication confidence is used as a state signal for adaptation and is not exposed as a numeric candidate-facing score. The model is not presented as a clinically or psychologically validated measure of confidence.

Table 4. Technical evaluator parameters in the tested configuration.

Parameter	Value	Purpose
S_1 / S_2 / R weights	0.15 / 0.35 / 0.50	Semantic, concept, and reasoning contributions
Concept threshold	0.30	Required-concept match cutoff
Reasoning floor	0.30	Activates concept-coverage dampening
Dampening factor	0.60	Reduces effective coverage when reasoning is weak
Bonus increment / threshold	0.03 / 0.50	Advanced-insight credit
Mistake increment / threshold	0.07 / 0.55	Misconception penalty
Penalty cap	0.30	Maximum accumulated penalty
Mandatory cap	0.60	Ceiling when a required concept is missing

7 Adaptive Difficulty and Bloom-Aware Question Selection

7.1 State, Action, and Difficulty

The current implementation uses a six-dimensional state consisting of current performance, rolling performance, communication confidence, hesitation, normalized session progress, and current difficulty. The state is normalized to $[0, 1]$. The PPO action set is

$$a_t \in \{\text{Easier}, \text{Same}, \text{Harder}\}, \tag{10}$$

with the corresponding difficulty update bounded to $D \in \{1, 2, 3, 4, 5\}$. The first two primary questions form the baseline phase; adaptive selection begins from Question 3. Follow-ups are subordinate turns and do not increase the primary-question counter.

7.2 Bloom Taxonomy as a Cognitive Difficulty Layer

Bloom's taxonomy [19, 20] provides a complementary interpretation of difficulty. The classic taxonomy and its revision organize learning objectives by increasing cognitive complexity. In PrepAIred, the project uses a five-band operational mapping from Understand through Create (Table 3) while retaining a separate five-tier numeric difficulty index for the adaptive policy. Thus the same technical topic can be represented at different cognitive levels: for example, explaining a known concept can occupy an Understand band, applying it to a code fragment an Apply band, diagnosing a failure an Analyze band, comparing design trade-offs an Evaluate band, and designing a new system an Create band.

This separation is useful for multi-agent coordination: the Question Selector owns item retrieval, the Strategy/PPO layer owns difficulty movement, and the

Bloom metadata remains a structural property of the candidate item rather than a quantity that an LLM is allowed to rewrite during the session.

7.3 Reward Scope and Evidence

The frozen PPO checkpoints were trained using a hybrid reward based on alignment to a hand-coded oracle, outcome delta, multimodal shaping, and stability. Because the training configuration uses a single simulated persona and unseeded candidate noise, the paper does not infer cross-candidate generalization from those runs. The coordination study instead evaluates the frozen policy as one proposal source inside a deterministic replay environment.

8 Agent Coordination, Arbitration, and Verification

8.1 Proposal Lifecycle

For each turn, the coordinator creates a versioned observation. Strategy and Coaching receive immutable views and return typed proposals. Each proposal passes four sequential checks: (1) strict schema validation with unknown fields rejected and NaN/out-of-range values refused; (2) forbidden-key scanning to prevent direct mutation of protected state; (3) optimistic concurrency verification against the committed turn version; and (4) grounding checks requiring proposed concepts to exist in evaluator evidence. An exception or timeout (150 ms in the tested configuration) drops the proposal and leaves the default policy active.

8.2 Arbitration Precedence

The deterministic precedence order is: (1) infrastructure constraints such as evaluator failure or difficulty bounds; (2) the guardrail layer, including rules that may prevent or force an action; (3) a validated Strategy proposal whose confidence exceeds $\tau = 0.6$ and which cites supporting evidence; and (4) the PPO action, or the labelled heuristic fallback if PPO is unavailable. A proposal that cannot take effect at the current boundary is recorded as modified. The coordinator remains the sole state writer.

8.3 Peer Messaging

The Strategy Agent may pass a typed peer message to Coaching. The message expresses pedagogical intent (for example, remediation or stretch mode) but does not expose private strategy state. In the tested configuration, this peer message changes coaching presentation rather than feeding back into Strategy's difficulty decision. This separation lets the paper test whether communication itself changes system trajectories rather than assuming that it does.

9 Experimental Methodology

9.1 Study A: Evaluator Evidence

Study A uses the 64-answer exploratory human-reference benchmark already present in the project records. The eight question clusters contain constructed technical answers; 54 answers use the mean of three recorded ratings, and 10 use recorded adjudicated values for cases where rater disagreement exceeded the working criterion. Rater provenance, formal independence, and institutional ethics status are incompletely documented; the benchmark is therefore not described as an expert-panel or confirmatory evaluation.

The three-rater inter-rater reliability was ICC(2,1)=0.953, but the benchmark is strongly separated by construction category. The paper therefore emphasizes cluster-aware inference and treats the results as evidence about this benchmark rather than as population validity.

For load-bearing statistics, the project uses a two-level bootstrap with seed 42 and $B = 10{,}000$: resample the eight questions with replacement and then resample the eight answers associated with each selected question. Paired signal contrasts use the same bootstrap draw for all compared signals.

9.2 Evidence Classes and Study B: Multi-Agent Coordination Replay

The coordination study uses the real InterviewOrchestrator in deterministic replay mode with evaluator outputs fixed. The replay corpus contains 64 technical-interview replay instances covering arrays, dynamic programming, trees/graphs, system design, concurrency, databases, networking, and object-oriented design. Each underlying replay cluster is combined with 20 seeded question-order permutations (seeds 42–61), yielding 160 sessions per condition. These 160 sessions are not independent; the eight underlying replay clusters are treated as the independent structural units in cluster-bootstrap analyses [18]. The evaluation distinguishes REPLAY, FAULT-INJECTED, SOCKET, and SYNTHETIC evidence classes. The coordination replay uses no live generative LLM; deterministic templates provide text output so that trajectory differences isolate coordination logic.

9.3 Baseline and Ablation Conditions

The primary baseline B0 is the original workflow without agents: frozen PPO (or fallback heuristic) plus guardrails and static follow-up thresholds. M1 is the full coordinated Strategy+Coaching system. M2 is the silent-agent condition. M3 removes Coaching. M4 removes arbitration checks. Under the Paper 2 labeling, M5 removes the grounding check, M6 removes Strategy-to-Coaching peer communication, and M7 disables PPO. Additional conditions compare canonical and deployed guardrail variants. The specialization question is evaluated by comparing the two-agent architecture against a unified controller that receives the same observations and uses equivalent decision thresholds.

9.4 Metrics

We report session completion and authority invariants; difficulty-level changes and reversals per session; action agreement with PPO; proactive follow-up turns; end-to-end coordination latency; agent calls per session; decision agreement between split and unified architectures; and sensitivity to arbitration threshold τ. All confidence intervals over replay effects are cluster-bootstrap percentile intervals across the eight replay clusters. They are descriptive rather than formal hypothesis tests.

10 Results

10.1 Evaluator Evidence

The available benchmark contains 64 constructed answers across eight technical questions. The CrossEncoder-only signal has Spearman $\rho = 0.4832$ with the human-reference score; using the two-level cluster-bootstrap analysis, the reported 95% CI is $[0.2453, 0.6956]$. The deployed composite reaches $\rho = 0.3812$ with 95% CI $[0.1529, 0.6473]$. A paired question-aware contrast between composite and R is -0.1020 with 95% CI $[-0.2853, 0.1171]$, so the current benchmark does not establish a reliable difference.

The word-count-only baseline reaches $\rho = 0.4897$. The CrossEncoder score is also associated with answer length ($\rho = 0.6594$, 95% CI $[0.457, 0.831]$). Controlling for length reduces the human-reference partial correlation to 0.2434 (95% CI $[-0.008, 0.507]$). Thus, the available evidence does not establish a length-independent reasoning contribution, particularly given that R contributes 50% of the unpenalized composite weighting.

The benchmark's three-rater reliability is high, but the items were constructed in well-separated answer-quality categories and rater provenance is incompletely documented. These results are therefore treated as exploratory component-validity evidence rather than population-level validation.

10.2 Coordination Dynamics

All 160 sessions in each tested condition completed, committed scores matched recorded evaluator outputs, and difficulty remained within $[1, 5]$ with one-level transition bounds. In the silent-agent condition M2, trajectories reproduced the baseline exactly, providing a useful transparency check.

Enabling the full coordinated system M1 changed the difficulty path in all 160 replay sessions. Mean level changes decreased from 15.119 to 12.45, a difference of -2.67 with cluster-bootstrap 95% CI $[-4.50, -0.24]$. Mean directional reversals decreased from 11.744 to 9.869, a difference of -1.88 with CI $[-3.12, -0.27]$. Proactive probing added $+0.59$ turns per session, with cluster CI $[0.00, 1.76]$; however, all of that additional probing came from one of the eight replay clusters, so the magnitude should not be generalized.

Table 5. Primary coordination results from deterministic replay (160 sessions per condition).

Metric	B0	M1	Observed change / interpretation
Level changes/session	15.119	12.450	-2.67; cluster CI $[-4.50, -0.24]$
Reversals/session	11.744	9.869	-1.88; cluster CI $[-3.12, -0.27]$
Proactive probes/session	0	+0.59	Cluster CI $[0.00, 1.76]$; driven by one cluster
Median turn latency	0.546 ms	1.589 ms	+1.04 ms coordination overhead
Agent calls/session	–	47.7	Coordination cost in the tested configuration
Silent-agent trajectory	baseline	identical	M2 demonstrates transparent infrastructure

The multi-agent layer increased median end-to-end turn latency by about 1.04 ms (95% CI $[1.00, 1.08]$ ms) and executed about 47.7 agent calls per session. This is a measurable coordination cost, not a claim of practical infeasibility.

10.3 Limits of Agent Specialization

The two-agent specialization and the matched unified controller produced identical decision outputs on all 1,280 replay turns. Likewise, removing Coaching did not alter the Strategy-driven difficulty trajectory because Coaching has no authority or information pathway into difficulty arbitration. These findings do not invalidate the multi-agent architecture; they show that, under deterministic rules and matched information, decomposition alone did not introduce a different decision policy. The measurable value is therefore architectural—state isolation, role separation, auditability, and bounded authority—rather than a demonstrated superiority of one controller over another.

10.4 Coordination Mechanism Sensitivity

The arbitration threshold was swept over $\tau \in \{0.0, 0.5, 0.6, 0.7, 0.8, 0.9\}$ while holding evidence checks and guardrails active. At $\tau = 0.0$, mean level changes collapsed to 2.225 and PPO agreement to 0.014, reproducing the behavior of removing arbitration. For $\tau = 0.5$–0.8, the observed trajectory remained at 12.45 changes and 9.869 reversals; at $\tau = 0.9$, the Strategy Agent was silenced and the trajectory returned to the baseline. The threshold is therefore a critical mechanism in the tested coordination envelope.

Removing PPO changed difficulty trajectories in only 4 of 160 sessions (2.5%) in the registered replay condition, revealing strong behavioral alignment between the Strategy heuristic and the deterministic fallback. This negative result is important: the paper does not claim that the learned PPO policy provides a large marginal contribution under these frozen replay conditions.

10.5 Peer Messaging and Unified Control

Removing the Strategy-to-Coaching peer message produced difficulty trajectories identical to M1 across all 160 sessions. The peer message affected coaching presentation—depth, remediation labels, and emphasis—but did not feed back into Strategy's difficulty decision. In the unified-controller comparison, split and unified architectures agreed on all 1,280 turns. Thus the experiment distinguishes *coordination infrastructure* from *decision diversity*: the former is present and measurable, while the latter is not demonstrated under the tested deterministic configuration.

10.6 Fault Injection and Safeguard Ablations (XB)

Fault-injection tests removed individual proposal safeguards. Without arbitration, 38 of 40 injected cases bypassed the guardrail boundary; with arbitration enabled, 0 of 40 did. When the grounding check was removed, 40 of 40 injected cases admitted ungrounded concept emphasis; with the grounding check enabled, 0 of 40 did. These results establish containment for the specific injected fault set, not universal safety. Well-formed but substantively wrong proposals remained undetected by structural validation, showing that schema, authority, version, and lexical grounding checks do not guarantee semantic correctness.

10.7 Threshold Decomposition (XC)

The arbitration threshold was swept over $\tau \in \{0.0, 0.5, 0.6, 0.7, 0.8, 0.9\}$ with other evidence checks and guardrails held active. At $\tau = 0.0$, mean level changes fell to 2.225 and PPO agreement to 0.014, reproducing the collapse observed when arbitration was removed. For $\tau = 0.5$–0.8, mean level changes remained 12.450 and PPO agreement 0.774. At $\tau = 0.9$, the Strategy Agent was silent and the trajectory returned to the B0 baseline. The realized confidence values were discrete (0.45, 0.70, and 0.80), so the plateau across 0.5–0.8 should not be interpreted as broad threshold robustness.

10.8 Grounding Probe (XD)

A separate grounding probe found that the lexical grounding check accepted unsupported paraphrased credit in 57/57 unambiguous cases. An earlier set of 17 apparent rejections was attributed to artifacts in the check rather than successful semantic grounding. The grounding layer should therefore be interpreted as a structural/lexical constraint, not as a validated semantic verifier.

10.9 Contradicted Predictions and Sensitivity

Two registered predictions in the coordination replay are directly contradicted. A9 predicted that disabling PPO would alter at least half of the session paths; only 4/160 sessions (2.5%) changed. A10 predicted that switching between canonical

and legacy guardrails would change at least one trajectory; none of the 160 replay paths changed. These outcomes indicate that the Strategy heuristic closely matched the fallback under the tested replay and that the replay did not reach the boundary states where the two guardrail implementations differ. The exact B6 wording is not restated here because the available paper versions use inconsistent prediction labels.

A cluster-sensitivity analysis further limits the interpretation of proactive probing: the reported +0.59 extra turns per session were driven entirely by one of the eight replay clusters; leaving that cluster out gives a difference of 0.00. The level-change interval $[-4.50, -0.24]$ barely excludes zero, so the direction is more stable than its practical magnitude is certain. The Strategy Agent proposed no "Harder" actions in replay, leaving that branch unexercised.

11 Discussion

The results support a careful interpretation of multi-agent design for adaptive technical-interview systems. The Strategy Agent's persistent weakness ledger changes the trajectory of difficulty transitions, smoothing repeated up/down oscillations relative to the frozen baseline. This is an algorithmic behavior change, but it is not a pedagogical outcome: deterministic replay cannot establish that smoother trajectories improve learning, confidence, retention, or interview performance.

The more important systems result is the explicit separation of authority. The evaluator owns numerical technical evidence; the PPO policy proposes difficulty; Strategy can propose changes but cannot write the difficulty state; Coaching can modify feedback presentation but not technical outcomes; and the LLM can generate language but cannot choose scores or difficulty. This makes conflicts observable and gives the coordinator a deterministic place to enforce hard constraints. The same property also explains why the architecture remains stable when agents are silent or invalid: proposals can be dropped without changing ownership of the session state.

The specialization comparison is deliberately negative. Splitting Strategy and Coaching did not change the matched decision trajectory relative to a unified controller with equivalent information. A reviewer might therefore ask why the split is necessary. The evidence supports an architectural answer, not a performance answer: the split isolates domain-specific state ledgers, makes responsibility explicit, enables independent audit records, and creates a surface on which coordination mechanisms can be experimentally isolated. At the same time, the split introduces measurable orchestration cost. The paper consequently avoids the stronger claim that multi-agent decomposition is intrinsically better.

The Bloom layer plays a different role. It does not make the agents autonomous; it gives the Question Selector a structured cognitive attribute with which to constrain candidate items. This lets adaptive difficulty operate over both a numerical tier and a stable cognitive-design attribute. Existing research has used Bloom taxonomy to classify or calibrate question difficulty; PrepAIred uses the

taxonomy as metadata in a live adaptive interview workflow rather than as a standalone classifier.

The evaluator results reinforce the same design philosophy. The technical score has transparent, reproducible structure, but the exploratory human benchmark is too small and confounded to support a strong claim about reasoning quality. The CrossEncoder component correlates with the human-reference score, yet the word-count baseline is comparable, and controlling for length removes much of the apparent association. The system therefore benefits from a protected deterministic evaluator architecture even while the scientific validation of one component remains incomplete. In a multi-agent context, this separation is important: the generative agent can be useful without being trusted with numerical authority.

12 Limitations and Threats to Validity

The main limitations are methodological and architectural.

- **Deterministic replay.** The 160-session coordination study reuses eight underlying replay clusters across question-order permutations. Cluster bootstrap reduces artificial precision, but eight clusters remain a small structural sample, and replay cannot capture dynamic responses from real candidates.
- **No live LLM in coordination experiments.** The replay study stubs the LLM gateway and uses deterministic template generation. Therefore, the measured coordination effects do not establish live LLM quality, hallucination rates, or human-facing coaching quality.
- **No learning-outcome study.** The work does not establish candidate learning gains, retention, interview placement success, or user satisfaction.
- **Agent specialization scope.** The Strategy and Coaching agents are rule- and ledger-based, not separately trained LLM agents. The findings therefore concern bounded weak agency and deterministic coordination, not open-ended emergent MAS behavior.
- **Evaluator validation.** The 64-answer benchmark is exploratory, internally constructed/recorded, and has incomplete rater-provenance and ethics documentation. Its eight-question cluster structure limits power for fine-grained comparisons.
- **CrossEncoder provenance.** The training data, exact split, and complete training code for the derivative CrossEncoder are not recoverable from the project record, so the paper reports measured behavior without claiming a specific learned reasoning property.
- **PPO evidence.** The frozen PPO evidence derives from a single simulated persona configuration with unseeded candidate noise; cross-persona generalization and superiority over simpler policies are not established.
- **Version-sensitive implementation details.** The recorded experiments depend on specific model-library behavior, guardrail variants, and configuration choices. The reported numerical results should therefore be interpreted as configuration-specific rather than universal properties of all software versions.

– **Confidence model.** Speech-confidence features are sensitive to acoustic conditions and are not presented as a validated psychological measure of confidence.

Data and reproducibility. The replay study is based on de-identified offline inputs and fixed software/model configurations. The current records do not provide complete rater provenance or a separate institutional ethics determination for the exploratory evaluator benchmark; no claim of population-level human validity is therefore made.

13 Conclusion

PrepAIred demonstrates how adaptive technical-interview practice can be organized as a centrally coordinated, bounded-authority multi-agent system rather than as an LLM-only interviewer. The architecture combines a deterministic technical evaluator, a separate communication-confidence signal, Bloom-aligned question metadata, a frozen PPO difficulty proposal, stateful Strategy and Coaching agents, typed coordination contracts, and a local generative model restricted to grounded language production.

The empirical findings are deliberately mixed. The coordinated agent layer changes difficulty trajectories and reduces oscillation in deterministic replay, but it adds orchestration overhead, and the matched split-versus-unified comparison produces identical decisions. The strongest interpretation is therefore architectural: explicit specialization provides isolated state ownership, proposal contracts, auditable arbitration, and fault containment, while the experiments do not establish that agent decomposition intrinsically improves decision quality. The evaluator benchmark reaches moderate agreement with human-reference scores but also shows that answer length and simple lexical signals explain much of the observed association, so no strong length-independent reasoning claim is made.

Future work should use new human-authored data to test the evaluator beyond length confounds; compare the specialization layer against stronger learned unified and multi-agent baselines; evaluate live LLM-backed agents with human participants; and investigate whether bounded authority remains beneficial under larger agent populations, richer peer communication, and decentralized coordination. These extensions preserve the central design principle of the current system: agent autonomy should grow only when the evidence and authority boundaries required for safe coordination grow with it.

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